CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 1: Rational and Irrational Numbers

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Insert two rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$.
2. Without dividing, state whether $\frac{13}{125}$ and $\frac{7}{30}$ are terminating or recurring.
3. Express $0.\overline{45}$ as a fraction in lowest terms.
4. State whether each is rational or irrational: $\sqrt{49}$, $\sqrt{12}$, $0.\overline{6}$, 2π.
5. Simplify $\sqrt{75} - \sqrt{48} + \sqrt{27}$.
6. Simplify $(\sqrt{5} + \sqrt{2})^2$.
7. Rationalise the denominator of $\frac{6}{\sqrt{7} - \sqrt{5}}$.
8. If $\frac{5 + 2\sqrt{3}}{7 + 4\sqrt{3}} = a + b\sqrt{3}$, find a and b.
9. If $x = 3 - 2\sqrt{2}$, find $x + \frac{1}{x}$.
10. Prove that $2\sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational.

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Answer Key

1. For example $\frac{7}{15}$ and $\frac{8}{15}$
2. $\frac{13}{125}$ terminates; $\frac{7}{30}$ recurs
3. $\frac{5}{11}$
4. Rational, irrational, rational, irrational
5. $5\sqrt{3} - 4\sqrt{3} + 3\sqrt{3} = 4\sqrt{3}$
6. $7 + 2\sqrt{10}$
7. $3(\sqrt{7} + \sqrt{5})$
8. $a = 11, b = -6$
9. 6
10. If $2\sqrt{3} = r$ were rational, then $\sqrt{3} = \frac{r}{2}$ would be rational, a contradiction

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